

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Python** data pipelines on **AWS Lambda** to process and prepare data for ML model training, handling approximately **3,500 events/sec** during peak hours.
- Developed and deployed scalable **Python** tools and services for annotated data inspection and bias investigation, integrating with **Databricks** for large-scale data processing.
- Built **data visualization dashboards** using **Power BI** to present model quality results and data inspection findings to data science and engineering teams, improving insight delivery speed by **~20%**.
- Applied software engineering practices to develop database designs and implementations in **PostgreSQL**, focusing on schema evolution and data integrity for ML feature stores.
- Collaborated with **data scientists** and **ML researchers** to understand their data requirements and translate them into robust data pipelines and engineering infrastructure.
- Managed CI/CD pipelines using **Github Actions** for automated testing and deployment of data engineering tools and services, reducing manual deployment time by **40%**.

SKILLS

Programming & Scripting – Python, SQL, R

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Databricks

Data Engineering & ML – Data Pipeline Design, ML Feature Engineering, Big Data Frameworks (Spark Concepts), Data Visualization, Database Design, CI/CD, API/SDK Interaction

Tools & Technologies – Power BI, Tableau, PostgreSQL, Github Actions, Git

Methodologies – Agile, Scrum

PROJECTS

ML Data Pipeline for Advertising Optimization

- Engineered a **Python**-based data pipeline on **AWS** to ingest and transform large volumes of advertising impression data, supporting downstream model training for brand safety classification.
- Implemented **SQL** queries and data transformations within the pipeline to ensure data quality and consistency, processing approximately **50GB** of raw data daily.
- Developed a **Power BI dashboard** to monitor pipeline performance and data quality metrics, providing actionable insights to the data science team for continuous improvement.

Production Data Quality Monitoring Service

- Built a **Python** service deployed on **AWS Lambda** that continuously monitors data quality for ML models in production, flagging anomalies and potential biases.
- Integrated the service with **PostgreSQL** to store historical data quality metrics and alert thresholds, enabling trend analysis and proactive issue resolution.
- Designed and implemented an alerting mechanism that notifies relevant teams via integrated channels when data quality deviates from established benchmarks, reducing critical data incidents by **15%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)